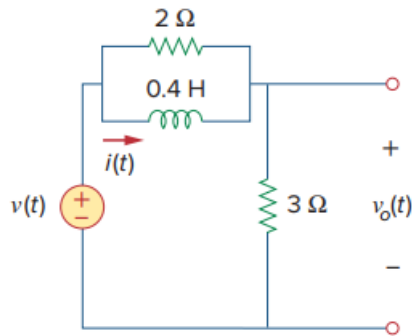


Problem

For the circuit in Fig. 7.98, determine $v_o(t)$ when $i(0) = 5$ A and $v(t) = 0$.

Fig. 7.98:



Step-by-step solution

Step 1 of 4

Refer to Figure 7.98 in the textbook.

Redraw the circuit diagram for $v(t) = 0$ V as shown in Figure 1.

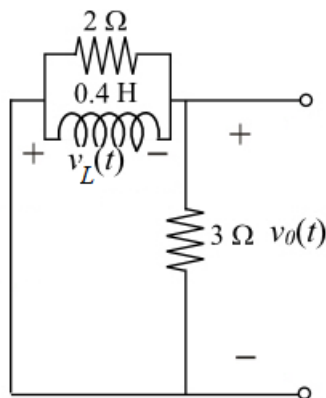


Figure 1

Step 2 of 4

Write the general expression for the inductor current for $t > 0$.

$$i(t) = i(0)e^{\frac{-t}{\tau}} \dots\dots (1)$$

Here,

τ is the time constant and is equal to the $\frac{L}{R}$.

$i(0)$ is the initial current flowing through the inductor.

Step 3 of 4

Calculate the equivalent resistance that appears across the open terminals of the inductor by short circuiting the voltage source.

$$\begin{aligned} R_{eq} &= 2 \parallel 3 \\ &= \frac{2(3)}{2+3} \\ &= \frac{6}{5} \Omega \end{aligned}$$

Calculate the time constant, τ for a RL circuit.

$$\tau = \frac{L}{R_{eq}}$$

0.4 H for L , $\frac{6}{5} \Omega$ for R .

$$\begin{aligned} \tau &= \frac{0.4}{\left(\frac{6}{5}\right)} \\ &= \frac{1}{3} \\ &= 0.33 \text{ s} \end{aligned}$$

Substitute 5 A for $i(0)$ and $\frac{1}{3}$ for τ in equation (1).

$$\begin{aligned} i(t) &= 5e^{-\left(\frac{t}{\left(\frac{1}{3}\right)}\right)} \\ &= 5e^{-3t} \text{ A} \end{aligned}$$

Thus, the current through inductor is $5e^{-3t} \text{ A}$.

Step 4 of 4

Calculate the voltage across the inductor.

$$\begin{aligned} v_L(t) &= L \frac{di(t)}{dt} \\ &= 0.4 \frac{d}{dt} (5e^{-3t}) \\ &= 0.4(5)(-3)e^{-3t} \\ &= -6e^{-3t} \text{ V} \end{aligned}$$

The voltage across inductor is $-6e^{-3t} \text{ V}$.

Apply KVL around the Loop in Figure 1.

$$\begin{aligned} v_L(t) + v_0(t) &= 0 \\ v_0(t) &= -v_L(t) \end{aligned}$$

Substitute $-6e^{-3t} \text{ V}$ for $v_L(t)$.

$$\begin{aligned} v_0(t) &= -(-6e^{-3t} \text{ V}) \\ &= 6e^{-3t} \text{ V} \end{aligned}$$

Therefore, the value of voltage $v_0(t)$ in the circuit is $6e^{-3t} \text{ V}$.